

Kutay Eroglu

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Istanbul, Turkey

OBJECTIVE

Computer Engineering Ph.D. student at [Bogazici University](#), specializing in self-supervised visual representation learning. Focused on joint-embedding predictive architectures, and biologically informed deep learning.

EDUCATION

- **Bogazici University** Sep 2026 - Present
Doctor of Philosophy in Computer Engineering Istanbul, Turkey
- **Bogazici University** Graduated Aug 2026
Master of Science in Computer Engineering Istanbul, Turkey
 - GPA: 3.57/4.00
- **Koc University** Graduated Jun 2022
Bachelor of Science in Computer Engineering Istanbul, Turkey
 - GPA: 3.07/4.00

EXPERIENCE

- **Bogazici University** Sep 2026 - Present
Graduate Research and Teaching Assistant Istanbul, Turkey
 - Assisting [CMPE354: Software Engineering](#), covering the software development lifecycle, engineering methodologies, and group project implementation.
- **Hitit** Oct 2022 - Sep 2025
Artificial Intelligence Engineer Istanbul, Turkey
 - Engineered and maintained [Pegasus Airlines'](#) marketing personalization stack, leveraging machine learning algorithms to enhance customer engagement and target 5 million customers per week.
 - Developed a flight delay prediction pipeline from scratch as a solo developer, covering feature engineering, modeling, continuous training, and product integration.
 - Led a team of 5 data scientists for 1 year, overseeing project delivery, Scrum events, agile board management, stakeholder communication, code reviews, and documentation.
- **Kariyer.net** Mar 2021 - Nov 2021
Backend Development Intern Istanbul, Turkey
 - Developed a one-time password mechanism in the [Isin Olsun](#) web application, handling creation of necessary API endpoints and database tables.
 - Created an API endpoint for internal customers that allowed querying of user login history.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] K. Eroglu and B. Gökberk (2026). **Noise-guided Masking for Image-based Joint-Embedding Predictive Architectures**. In *Proc. 34th IEEE Signal Processing and Communications Applications Conference (SIU)*. IEEE.
- [T.1] K. Eroglu (2026). **Noise-guided Masking for Image-based Joint-Embedding Predictive Architectures**. Master's thesis, Bogazici University.

PROJECTS

- **Noise-guided Masking for Joint-Embedding Predictive Architectures** Feb 2025 - Aug 2026
Master's thesis; I-JEPA, self-supervised learning IEEE SIU 2026
 - Proposed multi-noise masking, the first systematic study of noise-guided masking in a non-generative joint-embedding framework, integrating spectrally filtered noise maps with [I-JEPA](#) multi-block masking to guide patch removal inside context and target blocks.
 - On ImageNet-1K pre-training with ViT-B/16, improved the speed-accuracy trade-off in linear probing: 50.04% Top-1 accuracy vs. 49.1% for the multi-block baseline while reducing average pre-training iteration time by $\approx 26.8\%$.
 - Benchmarked against blockwise, rasterized, grid, and random masking baselines; ablations over patch removal ratio, context-block scale, backbone scale (ViT-S/B/L), and noise color showed consistent training speedups, with control experiments isolating intra-block patch removal as the source of gains.
- **Impact of an Anti-Biomimetic Training Regimen on Neural Network Organization and Behavior** Oct 2025 - Jan 2026
Term project; biologically inspired visual development Bogazici University

- Built on [NeurIPS 2024](#) workshop work on biomimetic training, introducing an anti-biomimetic regimen that transitions from high-resolution, full-color inputs to blurry, achromatic (grayscale) inputs in the second half of training.
- Compared standard (full-color throughout), biomimetic (blurry grayscale, then full-color), and anti-biomimetic (full-color, then blurry grayscale) regimens to isolate the role of temporal order in visual pathway organization.
- Reconfirmed that visual pathway organization is driven by the temporal order of sensory experience, not the data alone: late-introduced blur failed to induce a global shape bias (magnocellular-like), with networks instead reinforcing existing texture detectors under degradation.

• Delay Prediction

Jun 2024 - Sep 2025

Predicting probability of flight delays

Hitit

- Developed end-to-end development and production pipelines encompassing data preparation, feature engineering, model training, inference, and data export.
- Conducted a comprehensive literature review to evaluate existing modeling approaches and inform project strategy.
- Performed ablation studies to identify and eliminate redundant features and model components, improving performance and efficiency.
- Carried out exploratory data analysis (EDA) to uncover valuable features and determine the optimal modeling granularity.

GRANTS AND FUNDING

• TUBITAK EuroHPC National Access Call

2026

Project: Noise-guided Masking for Joint-Embedding Predictive Architectures

EuroHPC JU

- Awarded 256,000 GPU-hours on MareNostrum 5 (Barcelona Supercomputing Center) through the national access programme coordinated by TUBITAK (Scientific and Technological Research Council of Turkey).

ADDITIONAL INFORMATION

Languages: Turkish (Native), English (Professional Working Proficiency)